

NETWORK COMMUNICATION PROTOCOL

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1. Connection and Login

This flow documents the initial stages of coupling between the user and the application, highlighting the flexibility of the network provider to adopt two different transport technologies:

- **Network Initialization:** The application allows you to choose between RMI (by performing a lookup on the registry and obtaining a remote controller stub) and TCP (by establishing a socket connection and instantiating a client thread in the background dedicated to listening for packets).
- **Login Phase:** The user transmits their credentials to the Lobby component. If the authentication is successful and the necessary number of participants has been reached, the game match is initialized by informing the client appropriately; otherwise, an error message is returned.

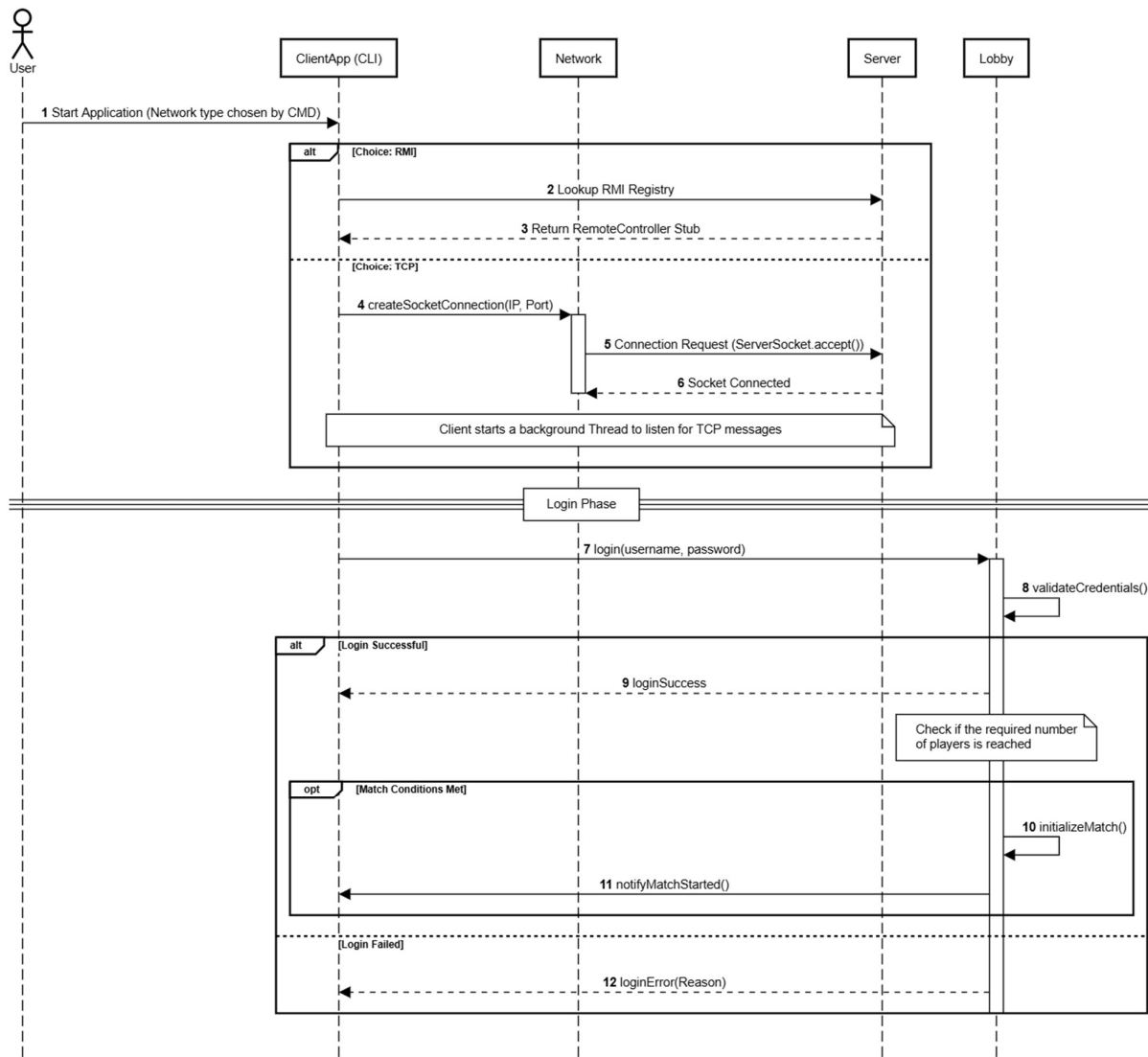


Figure 1: Architectural alternative between RMI/TCP and lobby login

2. Operations Management

The diagram describes the interaction between the Client (View) layer, the Network Layer and the Server-side components, divided into two distinct operational scenarios:

- Scenario 1 (Direct Operation): The client makes a request that travels from the network to the server. The response is notified directly to the client using the Observer pattern.
- Scenario 2 (Model Modification): The request traverses the controller to update the state of the business model. The variation of the data (Model) activates a Listener linked to an Adapter (ModelToRemoteViewAdapter), which finally takes care of asynchronously updating the remote view on the client.

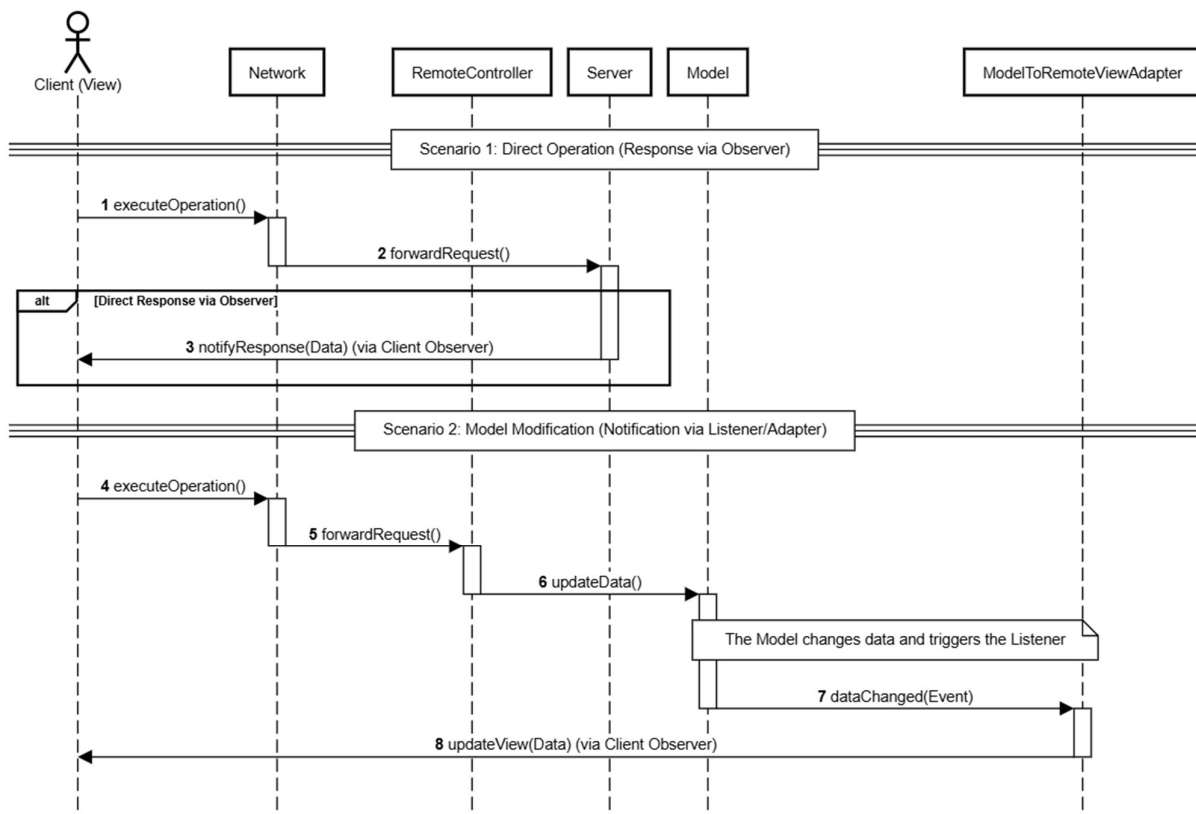


Figure 2: Direct execution flow and propagation of changes using the Adapter.

3. Network Status Monitoring

The last diagram illustrates the strategies adopted to ensure the robustness of the system and promptly intercept disconnections, differentiating them according to the chosen protocol:

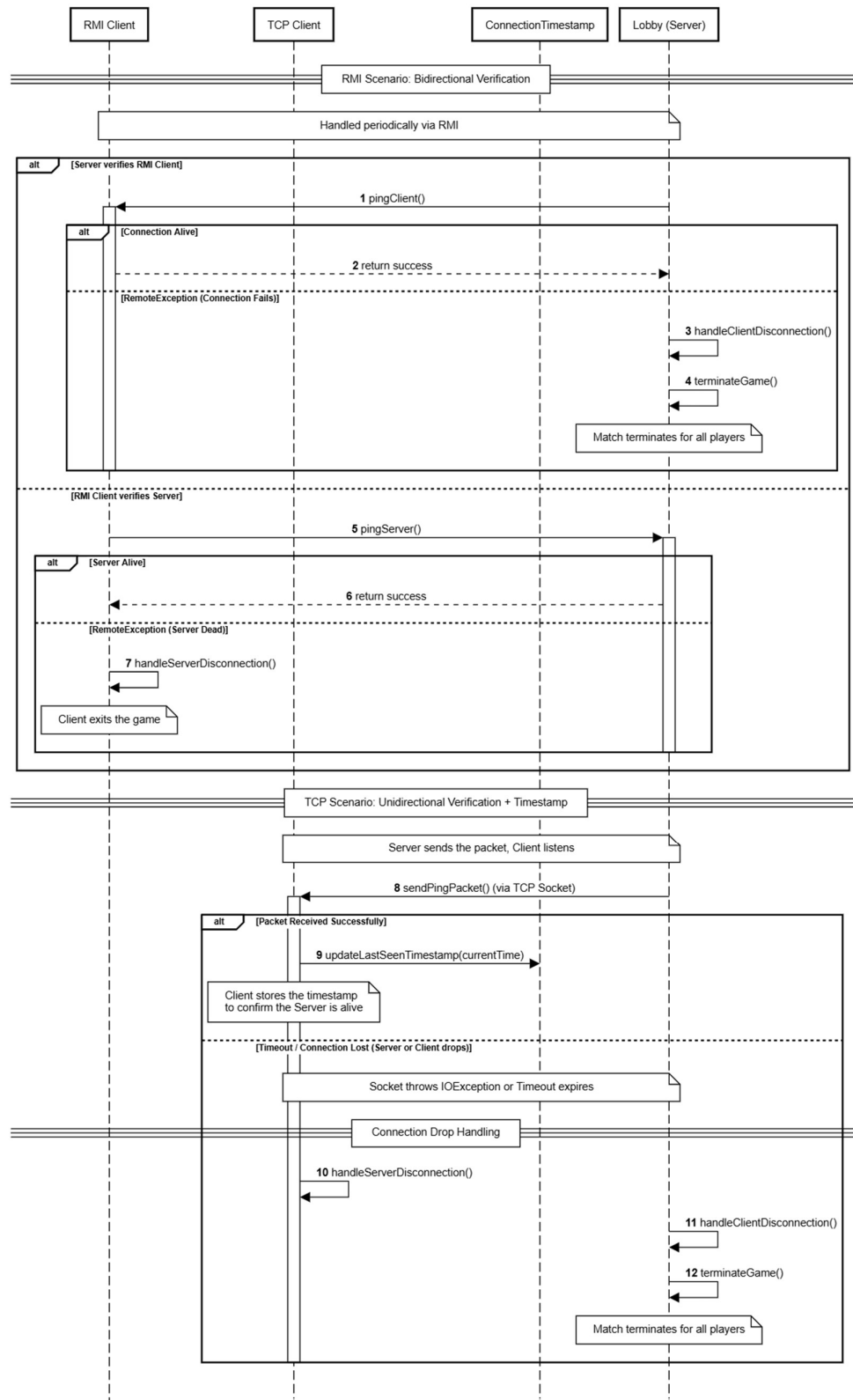


Figure 3: Heartbeat logic and exception handling for RMI and TCP.

- RMI (Two-Way Check) scenario: Server and Client periodically cross-check via ping() calls. The failure of a call generates a RemoteException that triggers the respective disconnection and termination routines of the match.
- TCP (One-Way Verification + Timestamp) Scenario: Server Sends Ping Packets Sockets. The Client intercepts the packet by updating a local timestamp. If an I/O timeout or exception occurs, the loss of the connection, leading to the correct closure of the game for all players involved.